Universidad LATINA de Panamá
SUMMUM DESIDERIUM SAPIENTIA

Cornell AAP
City and Regional Planning

Panama City, Panama
December 3rd , 2025

To: Linda Shi Foundation
921 University Ave,
213 W. Sibley Hall
Ithaca, NY 14853

Dear Professor Shi,

It is with great interest that I, Tomás Schmieder, a current student at Universidad Latina de Panamá, submit the attached proposal for the Climate Adaptation Lab in Panama City. The proposal is written to imagine how my university's existing community service mandate could be mobilized to support climate adaptation in my home city.

The project responds to a basic gap in Panamá's current adaptation efforts, we have national indicators and plans, but little neighborhood-scale data and almost no systematic way to incorporate community knowledge. The Climate Adaptation Lab proposes a one-year pilot that coordinates Trabajo Comunal Universitario students, municipal agencies, and residents in two high-risk neighborhoods to collect and use climate risk and social vulnerability data, drawing inspiration from NOAA and NASA's GLOBE program.

With support from the Linda Shi Foundation, this pilot could demonstrate a feasible model for turning existing institutional mandates into an evidence based and more equitable adaptation system. Thank you for considering this proposal.

Sincerely,

Tomás Schmieder

Introduction

Panama City has invested significantly in climate adaptation planning through *Panamá Resiliente* (2019) and a national Monitoring and Evaluation (M&E) system yet faces a fundamental problem: the city has created institutional frameworks without the operational capacity to make them function. The 2019 Resilience Strategy explicitly acknowledges that "local authorities do not have a full map of risks in the urban basins; it requires a comprehensive, validated tool to integrate the map of risks to the ordering of plans and thus foster an informed decision-making process" (p. 118). Additionally, a 2022 evaluation revealed that Panama's M&E system for climate change had apparent weaknesses, 52% of climate indicators lacked proper data collection specifications, 38% had misaligned targets, and only 2 of the 21 climate indicators met quality standards (ICAT, 2022). All 21 indicators rely on institutional data providers, government agencies, research institutions, and private companies, with zero indicators incorporating community-generated data. This gap between vision and implementation causes adaptation to become maladaptive. Panama's current approach privileges visionary narratives over operational tools for empirical monitoring, leaving the city vulnerable to investments that fail to address the needs of communities most affected by climate change. Addressing this issue requires establishing a collaborative data platform that links municipal agencies, universities, and communities, to transform adaptation from aspirational planning to transparent, evidence-based practice.

This proposal presents a specific intervention to achieve their transformation: the Climate Adaptation Lab (CAL). The CAL is designed as an evolution of existing institutional capacity. Article 301 of Panama's constitution mandates that university students provide 100 hours of community service before graduation, operationalized through Trabajo Comunal Universitario (TCU) programs that have functioned since 1991 at institutions like Universidad Latina de Panamá. By redirecting this constitutionally mandated service toward climate adaptation monitoring, the CAL addresses a critical capacity gap between Panama's adaptation plans and their implementation.

Theoretical Framework

The CAL builds on three interconnected theoretical insights from climate adaptation literature, concepts of maladaptation, environmental justice, and social vulnerability to environmental hazards.

First, Barnett and O'Neill's (2010) concept of maladaptation highlights the risks of action without evidence. Maladaptive interventions can "increase emissions of greenhouse gases, disproportionately burden the most vulnerable, have high opportunity costs, reduce incentives to adapt, and set paths that limit the choices available to future generations." Panama Resiliente's lack of spatial risk baselines and performance indicators exemplifies this risk, as without an integrated information base, the city risks funding initiatives that fail to aid vulnerable communities or even worsen their exposure. The CAL works to address this by creating the evidence base needed to prevent maladaptive investments.

The second framework is a warning from Anguelovski et al. (2020), where it is argued that urban greening is "grounded in technocratic principles and the naive apolitical assumption that greening will, unassisted result in both more just and prosperous cities." This warning acts to

express that urban planning can exclude social aspects from data production. Their framework of emancipatory adaptation requires repositioning marginalized communities not as service recipients but as knowledge producers whose firsthand experiences and understanding of climate impacts can work as evidence. Panama's current M&E system lacks mechanisms for communities to generate or verify environmental data, rendering marginalized communities invisible in decision-making processes. The CAL transforms this dynamic by working with local communities to produce data as a structural requirement of the monitoring system.

Third, the proposal draws on Cutter et al.'s (2003) work on social vulnerability to environmental hazards, emphasizing that risk is shaped not only by physical exposure to floods, landslides, or storms, but also by social conditions such as income, status, gender, age, and political marginalization. Panama's current M&E system treats communities primarily as recipients of risk information rather than as producers or validators of the data that define their vulnerability. Bringing communities into the monitoring process is therefore not only a technical fix for data gaps but also a way to address the invisibility of marginalized groups in adaptation governance.

Together, these frameworks reveal the CAL as addressing a form of infrastructural maladaptation, Panama has built monitoring systems without creating the workforce, processes, or participatory mechanisms needed to operate them. This produces the illusion of progress while not fixing the underlying issues of data gaps and exclusionary governance practices that heighten maladaptation risks.

Implementation Framework for the Climate Adaptation Lab

The core innovation of the CAL is recognizing that Panama already possesses the institutional foundation for community-engaged climate monitoring, but what is missing is coordination. Universidad Latina de Panama's TCU program has operated continuously since 1991 under the constitutional mandate of Article 301, which requires students to provide temporary services to the community before freely exercising their profession. The program requires 100 mandatory hours for all terminal degrees, coordinated by an established office with standardized approval procedures, reporting requirements, and community partner networks. TCU regulations already approve activities directly relevant to climate adaptation, conducting surveys, running community workshops, developing projects for vulnerable neighborhoods, and executing field research. However, this renewable workforce has never been systematically directed toward climate data collection for the vulnerability mapping and loss-and-damage mapping that adaptive governance requires.

In designing the CAL, this proposal adapts lessons from NOAA and NASA's GLOBE (Global Learning and Observations to Benefit the Environment), program, "focuses on advancing Earth systems science through data collection and analysis by volunteers" (GLOBE, 2025). This program mobilizes students and citizen scientists worldwide to collect standardized environmental measurements and feed them into an open, long-term database for Earth system monitoring. The CAL aims to create a place-based, justice-oriented analogue of this model for Panama City's climate adaptation needs, focusing on social vulnerability and urban risk rather than only physical variables.

This proposal requests support from the Linda Shi Foundation for a one-year pilot of the CAL in Panama City. The requested support would finance a small coordination team, training and materials for students and community participants, and the development of an open, digital platform for risk data storage and visualization. The purpose of the pilot is to demonstrate that a university-based, community-engaged monitoring model, adapted from NOAA and NASA's GLOBE program of student environmental observations, can significantly strengthen Panama's climate adaptation monitoring and reduce information inequities between central agencies and vulnerable neighborhoods.

The one-year pilot would unfold in three sequential phases:
Phase 1 (Months 1-3) Indicator Selection and Curriculum Design: During the first phase, CAL staff would build on the existing national indicator set and ICAT's SMART (Specific, Measurable, Achievable, Relevant, and Time-Bound) evaluation of adaptation indicators, they would select a small group of indicators where community-level data can make the most difference, such as "built areas in floodplains," which currently has limited geographic coverage, "people permanently displaced by climate events," an indicator that lacks data specifications, and "landslide exposure." Parallel to this, the CAL would co-develop a short training curriculum for TCU students and community participants that covers climate risk, social vulnerability, basic survey and mapping methods, and data use. Funding from the Linda Shi Foundation would support this process and stipends for communities who participate in this development.

Phase 2 (Months 4-9) Student Deployment and Community Data Collection: In the second phase, the Lab would recruit and train an initial cohort of TCU students from Universidad Latina de Panama to complete their 100-hour requirement through CAL projects. Under supervision from CAL staff and community advisors, student teams work in two strategically selected neighborhoods. These sites represent *Panama Resiliente's* priority areas: Alcalde Díaz, which faces both flooding and landslide vulnerability, and Juan Díaz, a place that experiences chronic flooding. These teams would work to implement standardized data collection protocols modeled on GLOBE's replicable field measurements, using household surveys, and participatory mapping of past and present vulnerabilities, infrastructure failures, and social conditions. The Linda Shi Foundation's support would cover the materials needed for training, local transportation, workshops for communities, and the development of a web interface for uploading, checking, and visualizing the data.

Phase 3 (Months 10-12) Data Integration, Interpretation, and Policy Engagement: In the final phase, the CAL would focus on quality control, integration of community generated data into municipal and national systems. CAL staff and Ministry of Environment's M&E team would use the pilot to data to help fill gaps in at least three national adaptation indicators and to produce updated neighborhood-scale risk profiles. These results would then be shared with community representatives, municipal planners, and national officials to identify where infrastructure is most vulnerable and requires repair.

Throughout the twelve-month project, the CAL pilot would explicitly follow the guiding principles of its theoretical framework. First, it prioritizes evidence-based adaptation to reduce maladaptation risks by generating the empirical baselines that Barnett and O'Neill argue are necessary to avoid harmful investments. Second, it follows Anguelovski et al.'s (2020) anti-

subordination, emancipatory environmental justice approach by repositioning marginalized communities as knowledge producers whose experiences help define adaptation priorities. Third, it centers social vulnerability, drawing on Cutter et al. (2003), by explicitly measuring not only exposure to physical hazards but also the social conditions that shape who is most at risk. These principles anchor the CAL as an institutional mechanism for transforming how adaptation knowledge is produced and used in Panama City.

Progress Tracking Through SMART Goals and Objectives

To ensure that this plan can be successfully implemented, three goals and their supporting objectives have been laid out using the SMART framework to measure the success of the project. Together, these goals translate the CAL from a conceptual intervention into a one-year pilot project that can be tracked, evaluated, and, if successful, scaled.

The first goal is to strengthen Panama's adaptation monitoring capacity through a university-based CAL. To reach this goal, two time-bound objectives are put in place, one is to train a cohort of TCU students in climate adaptation monitoring. By month 9 of the pilot, the CAL will have designed and delivered a short training program and a number of TCU students will have completed the training plus their 100 hours of CAL fieldwork. Success of this objective will be measured by training completion of TCU students and hours logged. The second objective is to improve the quality and coverage of key national adaptation indicators; by month 10, the CAL will have worked with the Ministry of Environment to identify and generate pilot datasets for at least three national adaptation indicators where community generated data can be added. Progress can be tracked by the number of indicators updated with SMART-compliant methodologies and the existence of new datasets in the CAL platform.

The second goal is to embed community knowledge and social vulnerability into adaptation planning. The first objective to achieve this goal is to establish collaboration mechanisms among municipal, academic, and community actors. By month 4, the CAL aims to formalize coordination between Panama's Ministry of the Environment, Universidad Latina, and community groups in Alcalde Díaz and Juan Díaz. To reach this, organizations will hold at least three quarterly meetings over the pilot year to document and share progress. The second objective is to co-produce neighborhood-scale vulnerability assessments with residents. By month 11, the CAL will have completed vulnerability assessment reports produced through coordinated work between university students and local communities. These assessments will identify both hazard exposures and key social vulnerability factors and success will be measured by their completion on schedule and by documented participation of at least 50 residents across surveys.

The third goal is to demonstrate that community-generated data can influence municipal and national adaptation decisions. This tests whether the CAL's outputs move beyond data collection to shape how Panama City and national agencies plan and prioritize adaptation investments. It is critical for demonstrating to the Linda Shi Foundation and future funders that the pilot project can lead to concrete policy-relevant outcomes. The first objective to achieve this goal is to integrate CAL data into formal planning and monitoring processes. By month 12, at least one municipal or national planning or monitoring document should explicitly reference or use CAL-

generated indicators, maps, or vulnerability profiles. The second objective is to secure an institutional pathway to sustain the CAL model beyond the pilot. By month 12, Universidad Latina de Panamá will have formally recognized CAL projects as a standing TCU track, the Ministry of Environment M&E team will have incorporated CAL-based community data collection protocols into its official guidance, and the Municipality of Panama will have designated a staff liaison and at least a minimal annual budget to maintain the data platform and community groups. Achievement will be documented through signed agreements between organizations and the granting of a budget.

Taken together, these goals and objectives provide a clear framework for tracking whether the pilot of the CAL using the Linda Shi Foundation's funding could create feasible changes in monitoring capacity, community participation, and the way adaptation decisions are made in Panama City.

Conclusion

The CAL achieves transformation not by creating new institutions but by changing relationships inside existing ones, directly addressing the forms of infrastructural maladaptation identified earlier. By training students with the Ministry of Environment's M&E team and sending them to work alongside residents in Alcalde Díaz and Juan Díaz, the Lab creates a feedback loop between data needs, community experience, and municipal decision-making. As Barnett and O'Neill (2010) argue, this kind of evidence-based learning process is essential to avoid harmful, path-dependent investments.

At the same time, the CAL advances the emancipatory, anti-subordination justice agenda described by Anguelovski et al. (2020) and centers the social vulnerability dimensions highlighted by Cutter et al. (2003). Community members are treated as co-researchers whose knowledge shapes indicator design and priorities, while data on income and exposure ensure that adaptation focuses on those most at risk, not just high-value areas.

In this way, the CAL offers a concrete pathway for shifting Panama City's adaptation trajectory away from infrastructural maladaptation and toward evidence-based, justice-centered governance. With support from the Linda Shi Foundation, the one-year pilot project can demonstrate that this model is feasible and worth institutionalizing across universities, agencies, and neighborhoods.

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